

March Madness - Written Report Review

The BC-AppliedAnalyticsProject group completed a comprehensive project on basketball games in the March Madness tournament, attempting to predict the probability that the lower-numbered seed wins the game. There are many aspects of their written review that deserve praise and a select few that could be improved.

PROBLEM

The idea for this project is well thought out and coincides well with the tournament in 2026. It was a brilliant idea to apply the model to the current March Madness tournament, and it kept the model relevant and interesting for stakeholders. Previous tournament data is interesting but seeing these predictions play out in real time brings a whole new level to this project.

DATA

Another aspect that deserves praise is the guard against potential data leakage. This group manipulated the dataset so that the model could only view information on a team that had happened **before the game in question**. This was very smart of the group, and not something immediately obvious until it was mentioned. If the model was able to see the team's final record on the season and previous results, for example, that could heavily influence the prediction for the next game.

MODELING

The modeling section of this report is well done. It was a great idea to use a logistic regression as a baseline model, and the more complex models were justified. The reasoning behind the selection of the random forest model was sound and provided great details in robust feature interactions and non-linear relationships.

One area to potentially improve is the performance metrics. The group does a fantastic job of laying out the various metrics including accuracy, precision, recall, f1, and area under the ROC curve but it is difficult to understand which are most important in consideration for the best model. There is also no context for why these metrics were selected, and a brief explanation on why these metrics changed across models would be helpful.

Model explainability was also a strength for this group. Efficiency metrics, turnover control, and relative team strength all contributed heavily to predictions, and this is perhaps not too surprising. The only comment here is to elaborate on “efficiency metrics”, this could mean a variety of game metrics from turnovers to field goal percentage, and it would be helpful to understand which are most important.

BIAS DETECTION AND MITIGATION

While the current bias detection and mitigation section is sufficient, it focuses mainly on bias in the model. One additional area to explore in bias consideration is the potential bias in the dataset itself, such as against under-reported teams. This group did a great job in removing predictors with a large proportion of missing data, but didn't focus as much on teams with missing predictors. Teams with missing data could be ones that exist outside the power conferences, and therefore don't have as much media attention to report stats. Proceeding without filling in this missing data could be a systematic bias that favors teams with high visibility and are likely high-ranking as well. This bias could result in a model that favors the higher seeds simply because they have more predictors available.

PRODUCTION

One section that we appreciated in the code and code output reports of this project was the idea to retrain the model so that upsets carry more weight. This is great for real-world applications such as sports betting or bracket pools where upsets are favored with greater rewards than a win of a higher seed over a lower seed. This feature shows a great understanding of a potential client base for this project, and how model parameters can be adjusted to accommodate the circumstances (such as betting lines) that might arise for the user. However, this commentary is completely absent from the team's written report. If sports betting is the main environment for this model (as denoted in the problem statement), this is a critical section that should be included in the accompanying written report.